

Your submitted solution is only **valid** if it contains the following paragraphs at the top of the program. Please *fill* in *your name* and *Neptun code*.

```
/*<NAME>, <NEPTUN>
```

```
This solution was prepared and submitted by the student stated above for the assignment of the Programming course. I declare that this solution is my own work. I have not copied or used third party solutions. I have not passed my solution to my classmates, neither made it public.
```

```
Students' regulation of Eötvös Loránd University (ELTE Regulations Vol. II. 74/C.§) states that as long as a student presents another student's work - or at least the significant part of it - as his/her own performance, it will count as a disciplinary fault. The most serious consequence of a disciplinary fault can be dismissal of the student from the University.*//
```

When submitting on TMS upload the program with the output enabled that only prints the result of the specific subtask e.g.: if you are solving a) do not include the outputs for tasks b), c), d)!

The Battle for Middle-earth

The Middle-earth is preparing for a great battle, with various factions amassing their forces. The Council of Elrond has started monitoring troop deployments to understand the balance of power. The deployment data is represented as a matrix where each cell contains an integer representing the troop strength of a specific region.



Input

The first line of input contains two integers, N ($1 \leq N \leq 50$) for the number of regions and M ($1 \leq M \leq 50$) for the number of troop deployments for each region. These two numbers define the columns (number of regions) and rows (number of soldier groups) of your matrix. The next N lines each contain M integers ($0 \leq \text{troop strength} \leq 1000$), representing the strength of the deployed troops in the region.

Write a program that solves the following tasks:

Tasks:

- Print the maximum troop strength across the entire matrix.
- Count the number of regions where at least one troop strength is an even number.
- For each row, print the smallest odd number. The input guarantees that each row contains at least one odd number.
- Print the number of rows that contain at least one troop strength value greater than 500. After that, list each region's index (1-based).

Output

Each task's output should be separated by new lines. Do *NOT* print the text in red; the red color is used here solely to illustrate and distinguish each subtask's output for clarity.

Example

Input

```
3 4
15 25 35 45
50 60 500 81
7 100 111 720
```

Output

```
Task a):
720
Task b):
2
Task c):
15 81 7
Task d):
2
2 3
```